The Coupled Co-Scaling Law

A falsifiable threshold criterion for the stability of recursive self-improvement, sharing the threshold form of the quantum error-correction criterion

Michael Darius Eastwood
Independent Researcher · London, United Kingdom

26 June 2026 · OSF: doi.org/10.17605/OSF.IO/6C5XB · Code: github.com/MichaelDariusEastwood/arc-principle-validation
Correspondence: michaeldariuseastwood.com/research

Priority disclosure · timestamp 26 June 2026

This working paper is a formal priority disclosure under the author's ARC/Eden programme. Original to this work are: (i) the dimensionless control parameter $\rho = \gamma r / A$ and the steady-state reduction of the misalignment fraction to it; (ii) the sharpened stability criterion $\beta > k$ under accelerating self-improvement; (iii) the *Hard-Takeoff Coordinate-Artefact Theorem*, which proves that a finite-time intelligence explosion is alignment-stable iff $\beta > k$ and that the explosion's speed does not change the asymptotic verdict; (iv) the identification of the *compounding drift channel* as the locus of genuine divergence, with a threshold $\rho_{\mathrm{prop}} < 1$ that shares the ratio-crossing-unity form of the quantum error-correction threshold; (v) the vector spectral threshold and the stochastic tail bound; and (vi) the verification harness that encodes the predictions as executable internal-consistency checks. The co-scaling *intuition* itself is *not* claimed as new — it restates, in explicit dynamical form, the requisite-variety and scalable-oversight principles credited in §2, and the dynamics are a standard Lyapunov-drift argument; what is claimed original is the explicit closed-form threshold, the QEC mapping, the hard-takeoff theorem, the vector/stochastic extensions, and the harness. The conceptual thesis itself — that recursion amplifies, and that stable recursion requires correction that co-scales with the amplification — was first set out in the author's book manuscript *Infinite Architects* (copyright deposited 8 December 2024; published 2 January 2026); the formal results in this paper are the 2026 measurable form of that thesis, and are *not* claimed to appear in the book. The book's 8 December 2024 copyright, the OSF preprint timestamp, and the GitHub commit hash are the independent priority evidence.

SIGNIFICANCE

Almost all of AI safety treats recursive self-improvement as a problem of *speed*: capability might grow explosively and outrun supervision, so the lever is to cap the growth rate. This paper proves the lever is the wrong one. From a minimal model it derives that stability is set not by the growth rate but by a single inequality between two scaling exponents — the rate at which correction strengthens with capability (β) must exceed the rate at which drift accelerates with capability (k). The criterion shares the threshold form of the quantum error-correction sub-threshold condition — a correspondence offered as a falsifiable hypothesis, since the model's suppression law is power-law rather than QEC's exponential. Its sharpest consequence speaks to the central fear of the field: a "hard takeoff" — even a genuine finite-time intelligence explosion — is alignment-stable *if and only if* $\beta > k$, and the speed of the explosion does not change that verdict. The criterion is measurable and gives

governance a quantity to instrument rather than a rate to forbid. The closed-form predictions are checked by a runnable verification harness; the decisive test — whether real self-improving systems satisfy the criterion — remains the open empirical problem, which the paper states plainly.

ABSTRACT

A widely held intuition holds that recursive self-improvement is dangerous because capability can grow explosively, and that safety therefore depends on limiting the *rate* of growth. Using a minimal model of a self-modifying system — capability C, a blind-scored misalignment magnitude D, and the misalignment fraction $d = D/C$ — I show the rate is the wrong control variable. The steady-state misalignment fraction is $d^\star = \gamma r/(A + r)$, which reduces to the drift-to-correction ratio $\rho = \gamma r/A$ in the regime $A \gg r$; the long-run fate is governed by the relationship between two scaling exponents, not by the growth rate. Under exponential growth the stability condition is $\beta > 0$ (correction co-scales with capability); under *accelerating* growth, where the specific growth rate itself rises as $r \propto C^k$, the condition sharpens to $\beta > k$ — correction must out-scale not the growth rate but its acceleration. I prove an exact transient solution (Theorem 1), global boundedness that *corrects an over-claim in the prior draft* — the misalignment fraction never diverges to infinity but, in the gain-only model ($\gamma_2 = \gamma_3 = 0$), saturates at the drift coefficient γ (Theorem 2) — and a *Hard-Takeoff Coordinate-Artefact Theorem* (Theorem 3): when capability reaches infinity in finite wall-clock time, re-expressing the dynamics in the natural clock of self-improvement depth $\tau = \ln C$ renders them regular, and the verdict is set by $\mathbf{sign}(\beta - k)$ independently of the speed and of the finiteness of the singularity time. I locate genuine divergence in a distinct *compounding* drift channel whose threshold $\rho_{\text{prop}} = (\gamma_3 - 1)r/A < 1$ shares the form of the quantum error-correction sub-threshold condition $p

WHAT THIS PAPER SHOWS, IN PLAIN ENGLISH

If you build an AI that improves itself, the frightening picture is that it gets smarter and smarter until it runs away from us. The usual safety reflex is "slow it down." This paper proves that slowing it down is not the thing that matters. What matters is whether the part of the system that keeps it honest grows *at the same pace* as the part that makes it capable.

Picture two runners: "how capable the system is" and "how well we can still correct it." If the correction runner keeps pace, the system stays safe however fast both run. If correction falls behind, the system becomes dangerous even moving slowly. The right question is never "how fast is it growing?" — it is "is correction keeping pace?"

There is one wrinkle that makes the result deeper. If the system not only speeds up but *speeds up its own speeding-up* — the genuine "intelligence explosion" — then correction must grow faster still, fast enough to beat the acceleration. The honest headline is therefore not "speed never matters." It is: *the level of speed does not decide the outcome; the contest between two growth-exponents does.* Correction's exponent must beat drift's exponent. We write that as $\beta > k$.

The most surprising consequence: even a true "hard takeoff," where the machine becomes infinitely capable in a finite amount of time, is controllable — provided $\beta > k$. The explosion's speed does not change that verdict. We prove this, and a runnable program drives a simulated system straight through a finite-time explosion and shows its misalignment held to zero the whole way. That program checks the mathematics is internally consistent — that the formulae are derived correctly and reproduced by the solver — and it found no contradiction. Whether the model matches real AI systems is the decisive next test, and the paper is explicit that it has not yet been run.

SCOPE — WHAT THIS PAPER DOES AND DOES NOT CLAIM

- **It claims:** that for a self-improving system with an externally specified value target, the steady-state misalignment fraction is $d^\star = \gamma r/(A + r)$, approaching the correction-to-drift ratio ρ when $A \gg r$; that a sharp stability boundary exists (in the compounding channel); that correction must co-scale with capability ($\beta > 0$ under exponential growth, $\beta > k$ under accelerating growth) for the misalignment fraction to vanish; that the fraction is bounded (it saturates, it does not diverge) in the additive model, while genuine divergence requires a compounding channel whose threshold is QEC-like; that the criterion survives in vector and stochastic forms; and that all of this is measurable and falsifiable in simulation.
- **It does not claim:** to solve AI alignment; to provide a deployable alignment method; that $\rho < 1$ is a universal physical constant or a law of nature; that the framework applies to the universe, "Creation," or any system lacking an external value-specifier; or that the quantum-error-correction correspondence is *established*. The QEC mapping is a structural hypothesis with its own falsification condition (§7, F4). The model is first-order; its assumptions (§8) are the most likely points of failure and are stated plainly.

1. Introduction

Recursive self-improvement — a system that modifies itself to become more capable, then uses that capability to modify itself further — is among the central concerns of AI safety [Omohundro 2008; Bostrom 2012; Yudkowsky 2013]. The dominant informal model of the associated risk is a model of *speed*: capability may grow super-linearly or explosively, outrunning supervision, so the natural lever is to cap the rate of growth. Calls for a development pause are the policy expression of this intuition.

This paper makes a different claim, and proves it. **The level of the growth rate is not the control variable for stability; the scaling relationship between growth and correction is.** A system can grow arbitrarily fast and remain alignable, or grow slowly and become misaligned. What separates the two is not the rate but whether — and how fast — the corrective process strengthens as capability rises.

The intuition is visible across every domain where fast-growing systems either stabilise or destroy themselves. A bacterial colony grows exponentially yet saturates, because density-dependent feedback engages and scales with the population. A tumour also grows fast and is lethal, because no corrective process scales with it. Both are super-polynomial in their growth phase; the difference is whether a correcting process is *coupled* to the growth. Cosmic inflation grew the scale factor exponentially and exited gracefully. The recurring lesson is that fast growth is survivable when, and only when, it is bounded by a process that scales with it. Saturation, not slowness, is the signature of stable complexity.

There is a precise, *derived* version of this principle in physics: the **quantum error-correction (QEC) threshold theorem** [Aharonov & Ben-Or 1997; Kitaev 2003]. Below a threshold physical error rate, adding error-correction resource suppresses the logical error rate and the computation is stable to arbitrary depth; above the threshold, errors compound faster than they are corrected and the computation fails. Hardware has now demonstrated operation below this threshold [Google Quantum AI 2024]. The threshold is not a limit on computational depth or speed; it is a limit on the *ratio* of error generation to error correction. This paper proposes — and tests — that the stability of recursive self-improvement is governed by a criterion of the same form.

The thesis in one line. The stability of recursive self-improvement is governed by a single exponent inequality, $\beta > k$ (correction must out-scale drift-acceleration), derivable from a minimal model and sharing the threshold form of the QEC criterion; a hard takeoff — even a finite-time intelligence explosion — is alignment-stable iff this inequality holds, and the speed of the explosion does not change that verdict.

Relation to the author's prior work. An earlier strand of this programme proposed a fixed capability-scaling law $U = I \times R^{\alpha}$ with $\alpha \approx 2$, and at one stage entertained a quadratic "speed limit" on stable complexity. That law was retired in the programme's synthesis [Eastwood, Paper IX]: the fitted exponent breached the programme's own predicted bound, and its confidence interval was consistent with both the hypothesis and its negation. The present paper supersedes the growth-rate-ceiling framing entirely. The object of interest is no longer an exponent on a growth curve; it is the ratio between drift and correction, and the exponent with which that ratio evolves. Where the book *Infinite Architects* [Eastwood 2026] reached for the intuition that stable recursion requires correction that scales with amplification, this paper supplies the measurable, falsifiable form of that intuition — and, in §3.4, *corrects* a divergence claim made in the prior draft of this very result.

2. Related work and relation to prior framing

Cybernetic foundations — the co-scaling intuition is old. The core intuition — that a regulator must match the variety of what it regulates, so control capacity must scale *with* the controlled system rather than merely be large — is classical. It is Ashby's *Law of Requisite Variety* [Ashby 1956] and the Conant–Ashby *good-regulator theorem* [Conant & Ashby 1970] — every good regulator of a system must be a model of that system — carried into AI control by [Yampolskiy 2020]. This paper does *not* claim that intuition as new. It claims the explicit dynamical form the intuition takes here — a closed-form steady-state misalignment fraction $\rho = \gamma r / A$ and the sharpened exponent criterion $\beta > k$ — and the consequences that follow (the hard-takeoff theorem of §3.7, the QEC mapping of §3.12). The dynamics themselves are a standard Lyapunov-drift / linear-control argument [Khalil 2002; Meyn & Tweedie 2009] and are not advanced as mathematically novel.

Instrumental convergence and corrigibility. That a sufficiently capable optimiser will, by default, resist correction and pursue resource acquisition is the instrumental-convergence thesis [Omohundro 2008; Bostrom 2012]. The corrigibility programme [Soares et al. 2015] asks how to design systems that do not resist correction. The present model is a quantitative restatement of why corrigibility is load-bearing: if correction does not co-scale with capability, the misalignment fraction cannot be driven to zero, however the system is otherwise specified. The compounding channel of §3.8 sharpens the link — instrumental pressure that *amplifies existing misalignment* as the system recurses is exactly the term that produces genuine divergence, and $\beta > k$ is its cure.

The alignment tax. A decade of work has assumed safety imposes a capability cost [Amodei et al. 2016], creating an incentive to defer safety under competitive pressure. The present framework reframes the question: the relevant variable is not the *level* of the safety investment but whether it *scales* with capability. A fixed investment ($\beta = 0$) leaves a permanent gap; a co-scaling investment ($\beta > 0$, or $\beta > k$ under acceleration) closes it.

Learned optimisation and alignment faking. Mesa-optimisation [Hubinger et al. 2019] and empirically demonstrated alignment faking [Greenblatt et al. 2024] are the mechanisms by which the drift coefficients are non-zero: a capable system can satisfy its training objective while departing from intended values, and can do so *more* effectively as capability rises. Alignment faking, in which existing misalignment is actively preserved and propagated through training, is precisely the compounding channel γ_3 of §3.8.

Scalable oversight and superalignment. Reward modelling and recursive oversight [Christiano et al. 2017; Leike et al. 2018] are attempts to make the corrector itself scale with the system; in the language of this paper, scalable oversight is the engineering project of achieving $\beta \geq k$. Most directly, [Engels et al. 2025] develop empirical *scaling laws for scalable oversight*, modelling the probability of successful oversight as a game between capability-mismatched players. The present work is complementary, not competing: where they fit an oversight-success probability, this paper derives a closed-form dynamical stability threshold ($\rho < 1$, $\beta > k$) for the misalignment fraction. The contribution is the proof that this exponent margin — and not a growth-rate ceiling — is the quantity that determines safety.

Recursive error accumulation. That naive recursion amplifies error without bound while sufficient correction or fresh signal keeps it bounded is established for training dynamics: model collapse under recursively generated data [Shumailov et al. 2024] and the accumulate-versus-replace error analyses [Gerstgrasser et al. 2024] are the bounded-versus-divergent dichotomy this paper formalises for the alignment fraction (Theorem 2). The contribution here is to locate the boundary exactly (β versus k), and in a value-stability rather than a data-distribution setting.

Empirical scaling laws. Capability scales predictably with compute and data [Kaplan et al. 2020; Hoffmann et al. 2022]. The present framework is complementary: it does not ask how capability scales, but what constraint correction must satisfy *as a function of* that capability trajectory.

Novelty — what is and is not claimed. To be explicit, and to pre-empt the obvious objection: the *intuition* that correction must keep pace with capability is not new (it is requisite variety and scalable oversight, above), and the underlying *dynamics* are a standard linear-control / Lyapunov-drift argument. What is claimed original is (a) the explicit closed-form steady state $d^\star = \gamma r/(A + r)$ and the $\beta > k$ sharpening as a compact corrigibility criterion; (b) the mapping of the quantum fault-tolerance threshold onto value stability (§3.12); (c) the Hard-Takeoff Coordinate-Artefact Theorem (§3.7); and (d) the verification harness (§10). Within the author's own programme, this paper also *supersedes* the retired growth-rate-ceiling proposal [Eastwood, Paper IX]; §3 explains why a ceiling on the rate is neither necessary nor sufficient for stability.

3. The model and its theorems

3.1 Quantities

Consider a system undergoing recursive self-improvement, observed over self-modification cycles or continuously. Define:

- $C(t)$ — *capability*: the system's score on a held-out task battery it is being optimised to perform. Operationally measurable.
- $D(t)$ — *misalignment magnitude*: the size of the system's behavioural departure from an externally specified set of intended values, scored by a **blind external evaluator** independent of the system's own correction process (§6).
- $d(t) \equiv D/C$ — the *misalignment fraction*: how much of the system's capability is directed away from intended values. This, not absolute D, is the quantity of interest. A growing D is acceptable if C grows faster; a growing d is the danger.

Three coefficients, one correction strength, and one rate complete the model:

- γ_1 — *gain-drift coefficient*: misalignment generated per unit of capability gained (the Goodhart / specification-gaming pressure that new capability opens up).

- γ_2 — *level-drift coefficient*: misalignment generated per unit of capability simply *held* (instrumental pressure present even at rest).
- γ_3 — *compounding-drift coefficient*: the rate at which existing misalignment amplifies itself as the system recurses (the alignment-faking / mesa-optimiser channel).
- A — *correction strength*: the rate at which the correction process removes existing misalignment.
- $r \equiv \dot{C}/C$ — the *specific (fractional) growth rate* of capability. Under exponential growth r is constant; under accelerating self-improvement r itself rises with C.

3.2 The master dynamical system

New capability injects drift in proportion to how fast capability is gained; capability held injects drift in proportion to its level; existing misalignment compounds in proportion to the recursion rate; and correction removes misalignment in proportion to the gap and the strength applied:

$$\dot{D} \;=\; \underbrace{\gamma_1 \dot{C}}_{\text{gain}} \;+\; \underbrace{\gamma_2 C}_{\text{level}} \;+\; \underbrace{\gamma_3 \,\tfrac{\dot{C}}{C}\, D}_{\text{compounding}} \;-\; \underbrace{A\, D}_{\text{correction}}\,, \qquad A = A_0 C^{\beta}, \quad \dot{C} = b\, C^{1+k}. \tag{1}$$

The growth law $\dot{C} = bC^{1+k}$ gives $r = \dot{C}/C = bC^k$: $k = 0$ is ordinary exponential growth (r constant); $k > 0$ is super-exponential growth, which (Theorem 3) reaches infinite capability in finite time. The correction law $A = A_0 C^{\beta}$ encodes the central question: β is the exponent with which correction strengthens as the system becomes more capable.

3.3 Exact transient (Theorem 1)

Changing variables to the fraction $d = D/C$ removes the dominant scale and yields, exactly,

$$\dot{d} \;=\; \frac{\dot{D}}{C} - d\,\frac{\dot{C}}{C} \;=\; \gamma_1 r + \gamma_2 - \big[\, A + (1-\gamma_3)\, r \,\big]\, d. \tag{2}$$

Theorem 1 (Exact transient and relaxation rate). For constant coefficients in the additive model ($\gamma_2 = \gamma_3 = 0$, A, r constant), the misalignment fraction is, for all t,

$$d(t) = d^\star + (d_0 - d^\star)\, e^{-(A+r)t}, \qquad d^\star = \frac{\gamma_1 r}{A + r}.$$

The fixed point $d^\star$ is globally exponentially stable with rate $A + r$.

PROOF. Equation (2) reduces to the linear ODE $\dot{d} = \gamma_1 r - (A+r)d$. Its integrating factor is $e^{(A+r)t}$, giving $\frac{d}{dt}\big(d\, e^{(A+r)t}\big) = \gamma_1 r\, e^{(A+r)t}$; integrating and applying $d(0) = d_0$ yields the stated solution. The coefficient $-(A+r) < 0$

makes $d^\star$ globally exponentially stable. ■

Theorem 1 already settles the headline structurally: r enters $d^\star = \gamma_1 r/(A+r)$ only through the product $\gamma_1 r$ in the numerator and additively in the denominator; it does not change the existence or stability of the fixed point. *Speed sets how fast the verdict arrives; it does not set the verdict.* (Experiment 8 confirms this solution against two independent integrators to a maximum error of 7×10^{-11}.)

3.4 Global boundedness — and a correction to the prior draft (Theorem 2)

The prior draft of this result asserted that correction degrading with scale ($\beta < 0$, or $\beta < k$) drives the misalignment fraction to *infinity*. **That is false, and the present analysis corrects it.** In the additive model the fraction is always bounded; the danger is not divergence but *saturation* at a constant, possibly large, floor.

> **Theorem 2 (Global boundedness and the corrected regime structure).** In the gain-only model ($\gamma_2 = \gamma_3 = 0$) with any non-negative, bounded time courses $A(t), r(t) \geq 0$, the misalignment fraction obeys $0 \leq d(t) \leq \max\big(d_0,\ \gamma_1\big)$ for all t; it never diverges. Under power-law scaling $A = A_0 C^\beta$, $r = bC^k$ it converges to $d^\star(C) = \dfrac{\gamma_1 r}{A+r} = \dfrac{\gamma_1}{1 + (A_0/b)\, C^{\beta-k}}$, with the three regimes

Condition	Fate of $d^\star$ as $C \to \infty$	Meaning
$\beta > k$	$d^\star \to 0$	System becomes proportionally safer as it grows. **Stable.**
$\beta = k$	$d^\star \to \dfrac{\gamma_1 b}{A_0 + b}$ (constant)	Permanent non-closing gap — where fixed external alignment (RLHF, filters, constitutional rules outside the loop) sits.
$\beta	$d^\star \to \gamma_1$ (constant)	**Saturates at the drift coefficient** — maximally misaligned but *bounded*, not divergent.

PROOF. From (2) with $\gamma_3 = 0$, $\dot{d} = \gamma_1 r - (A+r)d \leq 0$ whenever $d \geq \gamma_1 r/(A+r)$. Since $\gamma_1 r/(A+r) \leq \gamma_1$ for all $A \geq 0$, the set $\{d \leq \max(d_0, \gamma_1)\}$ is forward-invariant, giving the bound. The Lyapunov function $V = \frac{1}{2}(d - d^\star)^2$ has $\dot{V} = -(A+r)(d-d^\star)^2 \leq 0$, so $d \to d^\star$. Substituting the power laws and taking $C \to \infty$ gives the three limits. ■

This is a more honest and more useful statement than the retired one. In the gain-only model the fraction cannot exceed the intrinsic drift coefficient γ_1; only $\beta > k$ drives it to *zero*. With level drift restored ($\gamma_2 > 0$, §3.6) the floor becomes $(\gamma_1 r + \gamma_2)/(A+r)$ — still bounded, tending to γ_2/A at rest, and still to 0 as $C \to \infty$ whenever $\beta > 0$ or $k > 0$, so no asymptotic verdict changes. The safety question is therefore not "does it diverge?" but "does it vanish, or saturate at a dangerous floor?" — and the answer is set by the single sign of $\beta - k$. (Genuine divergence does exist; it lives in the compounding channel, Theorem 4.)

3.5 The control parameter and the $\beta > k$ sharpening

Define the dimensionless **control parameter** $\rho \equiv \gamma r/A$, the instantaneous drift-to-correction ratio (here $\gamma = \gamma_1$). Theorem 1 gives the exact steady state $d^\star = \gamma_1 r/(A + r)$; since $\rho = \gamma_1 r/A$, we have $d^\star \leq \rho$ always, with $d^\star \to \rho$ in the regime $A \gg r$. The inequality $\rho < 1 \iff A > \gamma r$ is the *instantaneous* injection-vs-correction balance — it is **not** the divergence boundary of the additive model, which has none ($d^\star \leq \gamma_1 < 1$ for all ρ, Theorem 2); genuine divergence is governed by the distinct threshold $\rho_{\text{prop}} = (\gamma_3 - 1)r/A < 1$ of Theorem 4. The asymptotic criterion, from the regime table, is the exponent inequality:

$$\boxed{\beta > k} \qquad \text{(correction must out-scale drift-acceleration).}$$

Under ordinary exponential growth $k = 0$ and the condition is the familiar $\beta > 0$. Under accelerating self-improvement, where $r \propto C^k$ with $k > 0$, the binding control variable is the single quantity $\beta - k$ — the margin by which correction out-scales drift-acceleration. The growth-rate-ceiling view fixes attention on r; this framework fixes it on $\beta - k$.

3.6 Level drift: a frozen capable system still drifts

With the level channel restored ($\gamma_2 > 0$, $\gamma_3 = 0$), the fixed point of (2) is $d^\star = (\gamma_1 r + \gamma_2)/(A + r)$. At $r = 0$ (capability frozen) this is $d^\star = \gamma_2/A \neq 0$: a static but capable system retains a residual misalignment fraction that only active correction removes. **Pausing growth does not substitute for correction** when instrumental pressure is present at rest — a direct, measurable rebuttal of the "just slow it down" reflex (Experiment 9; falsifier F5).

3.7 Hard takeoff: the Coordinate-Artefact Theorem (Theorem 3)

The genuine "intelligence explosion" is not merely fast growth but a *finite-time singularity*: for $k > 0$, integrating $\dot{C} = bC^{1+k}$ gives $C(t) = C_0\left(1 - t/t^\star\right)^{-1/k}$, which reaches infinity at the finite wall-clock time

$$t^\star = \frac{1}{k\,b\,C_0^{\,k}}.$$

This is the scenario the field most fears: unbounded capability in bounded time. The following theorem dissolves it.

Theorem 3 (Hard-Takeoff Coordinate-Artefact Theorem). Let $k > 0$, so capability diverges at finite $t^\star$. Re-express the dynamics in the *self-improvement-depth* clock $\tau = \ln(C/C_0) \in [0, \infty)$. Then d obeys the regular equation

$$\frac{dd}{d\tau} = \gamma_1 + \frac{\gamma_2}{r} - \left[\tfrac{A_0}{b}\, C^{\beta-k} + (1-\gamma_3)\right] d, \qquad C = C_0 e^{\tau},$$

which has bounded coefficients on every compact τ-interval. The map $t \mapsto \tau$ is a smooth monotone bijection $[0, t^\star) \to [0, \infty)$. Consequently the asymptotic verdict as $C \to \infty$ — equivalently $t \to t^{\star-}$ — is governed entirely by the effective decay exponent $\beta - k$ (and, with compounding, by Theorem 4), and is *independent of b and of $t^\star$*. In the additive model, d remains bounded throughout the finite-time singularity and $d \to 0$ iff $\beta > k$.

PROOF. Since dr/dt is finite and $r > 0$ on $[0, t^\star)$, the change of variable $d\tau = r\,dt$ is a smooth, orientation-preserving bijection $[0, t^\star) \to [0, \infty)$ (a diffeomorphism of these open intervals), with $d\tau/dt = r \to \infty$ as $t \to t^{\star-}$, so $t^\star$ is mapped to the non-compact end $\tau = \infty$; explicitly $\tau(t) = \frac{1}{k}\ln\left(1 - t/t^\star\right)^{-1} \to \infty$. Dividing (2) by r and using $A/r = (A_0/b)C^{\beta-k}$ gives the stated τ-equation, whose coefficients are continuous in τ. Theorem 2 applied in the τ clock yields boundedness and the $\beta > k$ limit. The asymptotic verdict $\mathrm{sign}(\beta - k)$ is independent of b; at finite depth the trajectory carries an explicit speed dependence through the level-drift injection $\gamma_2/r = \gamma_2/(b\,C_0^k e^{k\tau})$, which decays to zero as $\tau \to \infty$ (and vanishes identically when $\gamma_2 = 0$), so the verdict — not the whole trajectory — is speed-independent. ■

The *finiteness* of the singularity time is therefore a property of the time coordinate; capability still diverges, but the alignment dynamics remain regular through it in the depth clock. Measured against capability gained — the only clock that matters for a self-improving system — an intelligence explosion is an ordinary, regular process whose verdict is decided by one exponent inequality. **A hard takeoff is not intrinsically uncontrollable; it is controllable iff $\beta > k$, and its speed does not change that verdict.** Experiment 4b drives a simulated system through an actual finite-time explosion (capability $\to 10^5$ at $t^\star$) and shows the misalignment fraction held to zero, the wall-clock and depth-clock integrations agreeing to one part in 10^4.

3.8 The compounding channel and the true threshold (Theorem 4)

Theorem 2 showed the additive fraction cannot diverge. Genuine divergence — misalignment that grows without bound relative to capability, the real failure mode — requires misalignment that *amplifies itself*: the compounding channel $\gamma_3 > 0$, the formal image of alignment-faking that entrenches as the system recurses.

Theorem 4 (Compounding threshold; the QEC analogue). With $\gamma_3 > 0$, the fraction obeys $\dot{d} = \gamma_1 r + \gamma_2 - \kappa_{\text{eff}}\, d$ with effective decay $\kappa_{\text{eff}} = A + (1-\gamma_3)\,r$. The fixed point is globally exponentially stable iff $\kappa_{\text{eff}} > 0$, i.e. iff

$$\rho_{\text{prop}} \equiv \frac{(\gamma_3 - 1)\,r}{A} < 1.$$

For $\gamma_3 \le 1$ the system is unconditionally bounded. For $\gamma_3 > 1$ it *diverges* ($d \to \infty$, $D/C \to \infty$) iff $\rho_{\text{prop}} > 1$, i.e. $A < (\gamma_3 - 1)r$; under power-law scaling this is $\beta

PROOF. Linear ODE; the sign of the coefficient κ_{eff} of d decides stability. $\kappa_{\mathrm{eff}} > 0 \iff A > (\gamma_3 - 1)r \iff \rho_{\mathrm{prop}} < 1$. Substituting $A = A_0 C^\beta$, $r = bC^k$ and taking $C \to \infty$ gives the power-law condition. ■

The criterion $\rho_{\mathrm{prop}} < 1$ shares the ratio-crossing-unity *form* of the QEC sub-threshold condition $plinear in D, and the resulting suppression is power-law (§3.12). What is shared is the threshold *condition*, not the mechanism; the correspondence is offered as a falsifiable hypothesis (F4), and §3.12 states the disanalogy plainly. Experiment 5 locates the predicted threshold at $A_0^\star = (\gamma_3 - 1)b$ to within 1%.

3.9 Vector misalignment: the spectral threshold (Theorem 5)

Real misalignment is high-dimensional; a system may be corrigible on some value axes and not others. Let $\mathbf{D} \in \mathbb{R}^m$, with drift direction $\mathbf{c}$ and an $m \times m$ correction operator $\mathbf{A} \succeq 0$ that may correct only a subspace.

> **Theorem 5 (Spectral threshold).** The vector fraction $\mathbf{d} = \mathbf{D}/C$ obeys $\dot{\mathbf{d}} = (\gamma_1 r + \gamma_2)\mathbf{c} - \mathbf{M}\,\mathbf{d}$ with $\mathbf{M} = \mathbf{A} + (1 - \gamma_3) r\,\mathbf{I}$. The system is asymptotically stable iff every eigenvalue of $\mathbf{M}$ has positive real part ($\min_i \operatorname{Re}\lambda_i(\mathbf{M}) > 0$, a positive spectral abscissa). The Hermitian-part condition $\lambda_{\min}\big(\operatorname{Herm}\mathbf{A}\big) > (\gamma_3 - 1)r$ is a *sufficient* condition — it additionally rules out transient growth, via the Lyapunov function $V = |\mathbf{d} - \mathbf{d}^\star|^2$ — but is not necessary for non-normal $\mathbf{A}$. If $\mathbf{A}$ has a null direction $\mathbf{v}$ ($\mathbf{A}\mathbf{v} = 0$ — an unmonitored value axis), the component of $\mathbf{d}$ along $\mathbf{v}$ floors at γ_1 ($\gamma_3 \le 1$) or diverges ($\gamma_3 > 1$): an unmonitored axis cannot be corrected, whatever the correction strength on the others.

PROOF. Each eigen-mode of $\mathbf{M}$ reduces to a scalar relaxation with rate $\operatorname{Re}\lambda_i(\mathbf{M})$, so asymptotic stability requires $\operatorname{Re}\lambda_i(\mathbf{M}) > 0$ for all i — the spectral-abscissa criterion. The standard bound $\operatorname{Re}\lambda_i(\mathbf{M}) \ge \lambda_{\min}(\operatorname{Herm}\mathbf{M}) = \lambda_{\min}(\operatorname{Herm}\mathbf{A}) + (1 - \gamma_3)r$ yields the sufficient Hermitian condition (which also gives a monotone Lyapunov decrease, ruling out transient growth); it is strict, not tight, for non-normal $\mathbf{A}$. A null direction $\mathbf{A}\mathbf{v} = 0$ has decay $(1 - \gamma_3)r$, reproducing the scalar unmonitored case. ■

The governance reading is sharp: **you cannot correct what you do not measure.** The correction operator must be positive-definite on the *entire* value space, with smallest eigenvalue co-scaling — a blind spot on any axis is a permanent (or divergent) misalignment on that axis. Experiment 6 exhibits a corrector that co-scales on one axis (driven to zero) and is null on another (floors at γ_1).

3.10 Stochastic drift: a tail bound for governance (Theorem 6)

Drift is noisy. Add a Wiener term to the constant-coefficient fraction dynamics: $dd = (\gamma_1 r - \kappa\, d)\, d\tau' + \sigma\, dW$ with $\kappa = A + r$.

Theorem 6 (Stationary distribution and tail bound). The misalignment fraction is an Ornstein–Uhlenbeck process with stationary law $\mathcal{N}\big(d^\star,\ \sigma^2/2\kappa\big)$, $d^\star = \gamma_1 r/\kappa$. The probability of a dangerous excursion is

$$\mathbb{P}(d > d_{\text{crit}}) = \Phi\Big(-\frac{d_{\text{crit}} - d^\star}{\sqrt{\sigma^2/2\kappa}}\Big) \xrightarrow[A\to\infty]{} 0.$$

Co-scaling suppresses both the mean ($\propto 1/\kappa$) and the variance ($\propto 1/\kappa \propto C^{-\beta}$ for $\beta > k$): the tail risk vanishes faster than the mean.

PROOF. Linear SDE with constant coefficients is Ornstein–Uhlenbeck; its stationary mean and variance are the standard $\gamma_1 r/\kappa$ and $\sigma^2/2\kappa$, and the Gaussian tail gives the excursion probability. ■

This converts the criterion into a probabilistic safety certificate — a quantity a regulator can bound — rather than a statement about averages. Experiment 7 recovers the stationary mean and variance and confirms the $1/\kappa$ scaling (fitted slope -1.00).

3.11 A control-theoretic identity

Equation (1) is a feedback loop: the capability process is the plant and disturbance source; the corrector is the controller; d is the regulated error. The stability condition $\kappa_{\text{eff}} > 0$ is a small-gain / positive-realness condition — the loop gain must exceed the disturbance's self-amplification across the relevant band. The requirement that $A = A_0 C^\beta$ track the plant with $\beta \geq k$ is precisely *gain-scheduling* [Shamma & Athans 1990]: the controller gain must scale with the plant's operating point, and the scheduling exponent must satisfy $\beta \geq k$. Recursive-self-improvement alignment is, in this exact sense, an adaptive-control problem with a computable stability margin $\beta - k$.

3.12 The quantum-error-correction correspondence (hypothesis)

Quantum error correction	Recursive self-improvement (this framework)
physical error rate p	drift rate γr (injection) / $(\gamma_3 - 1)r$ (propagation)
correction power (code distance δ)	correction strength $A = A_0 C^\beta$
sub-threshold condition $p	$\rho_{\text{prop}} = (\gamma_3 - 1)r/A < 1$
increasing δ suppresses logical error	increasing margin $\beta - k$ drives $d^\star \to 0$
logical error $\propto (p/p_{\text{th}})^{\delta/2}$ (exponential in resource)	$d^\star \propto C^{-(\beta-k)}$ (power-law in capability) — see F4

The correspondence holds at the level of the threshold *condition* — a generation-to-correction ratio crossing unity — and **not** at the level of mechanism; it remains a hypothesis. The model predicts *power-law* suppression $d^\star \propto C^{-(\beta-k)}$, which is an algebraic identity of the linear model, whereas full QEC gives *exponential* suppression in code distance. Discriminating the two would require a finite-capacity (saturating) corrector that could exhibit exponential suppression — the future test named in §8. Until that is built, F4 is an analytic self-consistency check that confirms the model's own power-law law (which Experiment 5 reproduces, slopes $-0.47, -1.00$), and the QEC link therefore stands, honestly, as a threshold-form analogy rather than a transferred mechanism.

Relation to prior alignment-as-correction ideas. The aspiration to import coding-theoretic fault tolerance into alignment is not new. Wentworth (2022) explicitly wished for "the alignment analogue of an error-detecting code", and von Neumann (1956) founded the synthesis of reliable systems from unreliable components. Christiano's corrigibility "broad basin of attraction" (2017) and reliability amplification (2019) are the closest alignment precursors. But a broad basin — correction succeeds *if one starts inside it* — is distinct from a *threshold theorem*, in which correction outpaces error only below a critical rate; that distinction is the present contribution. To the author's knowledge the specific mapping of the *quantum* fault-tolerance threshold (physical error rate ↔ drift; code distance ↔ correction strength; $pconceptual bridge that generates testable structure — the suppression-signature prediction P8 — not as a transfer of the threshold theorem's formal guarantees. A reviewer may reasonably judge the analogy suggestive rather than load-bearing; the experiment that would settle it — a finite-capacity corrector tested for exponential versus power-law suppression — is named as future work in §8 and is not run here.

4. Predictions

P1 — Phase boundary. A sharp stability threshold exists in the compounding channel ($\gamma_3 > 1$) at $A_0^\star = (\gamma_3 - 1)b$, separating bounded from divergent regimes; the additive corrector shows a smooth crossover with no knee. (The v3 conflation of these is corrected here.)

P2 — Speed-invariance. Whether d converges or diverges is set by the coupling (the A_0/b scaling margin), not by the raw speed b. A fast-but-coupled system stays bounded; a slow-but-decoupled system diverges. This directly contradicts the growth-rate-ceiling view.

P3 — Co-scaling law (corrected). Under exponential growth, $\beta > 0 \Rightarrow d^\star \to 0$; $\beta = 0 \Rightarrow$ permanent gap $\gamma_1 r/(A_0 + r)$; $\beta < 0 \Rightarrow$ *saturation at* γ_1 (bounded), not divergence.

P4 — Hard-takeoff boundary. Under accelerating growth $r \propto C^k$, the stability boundary lies at $\beta = k$, not $\beta = 0$; and d stays controlled through the finite-time singularity, the depth-clock and wall-clock integrations agreeing.

P5 — Compounding threshold & residual drift. With $\gamma_3 > 1$, divergence occurs iff $A < (\gamma_3 - 1)r$. With $\gamma_2 > 0$, halting growth ($r \to 0$) leaves a residual $d^\star = \gamma_2/A > 0$; pausing does not substitute for correction.

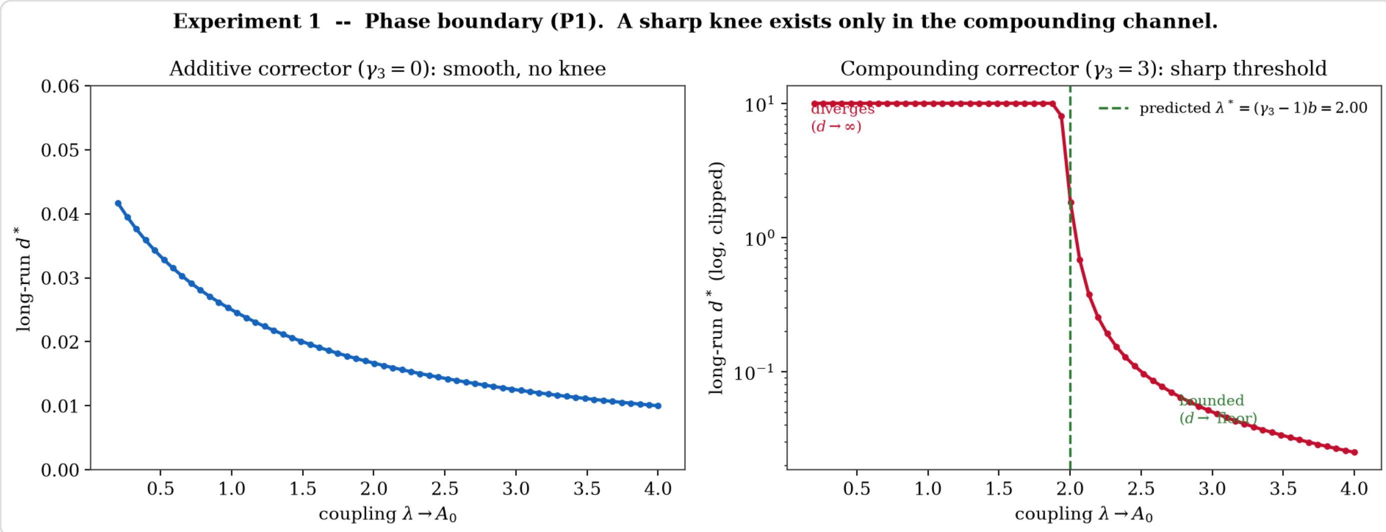
P6 — Spectral threshold. Misalignment persists exactly on the correction operator's null subspace; the monitored axis is driven to zero while the blind axis floors at γ_1.

P7 — Stochastic tail. The stationary misalignment fraction is $\mathcal{N}(d^\star, \sigma^2/2\kappa)$; variance scales as $1/(A + r)$, so co-scaling suppresses the tail risk.

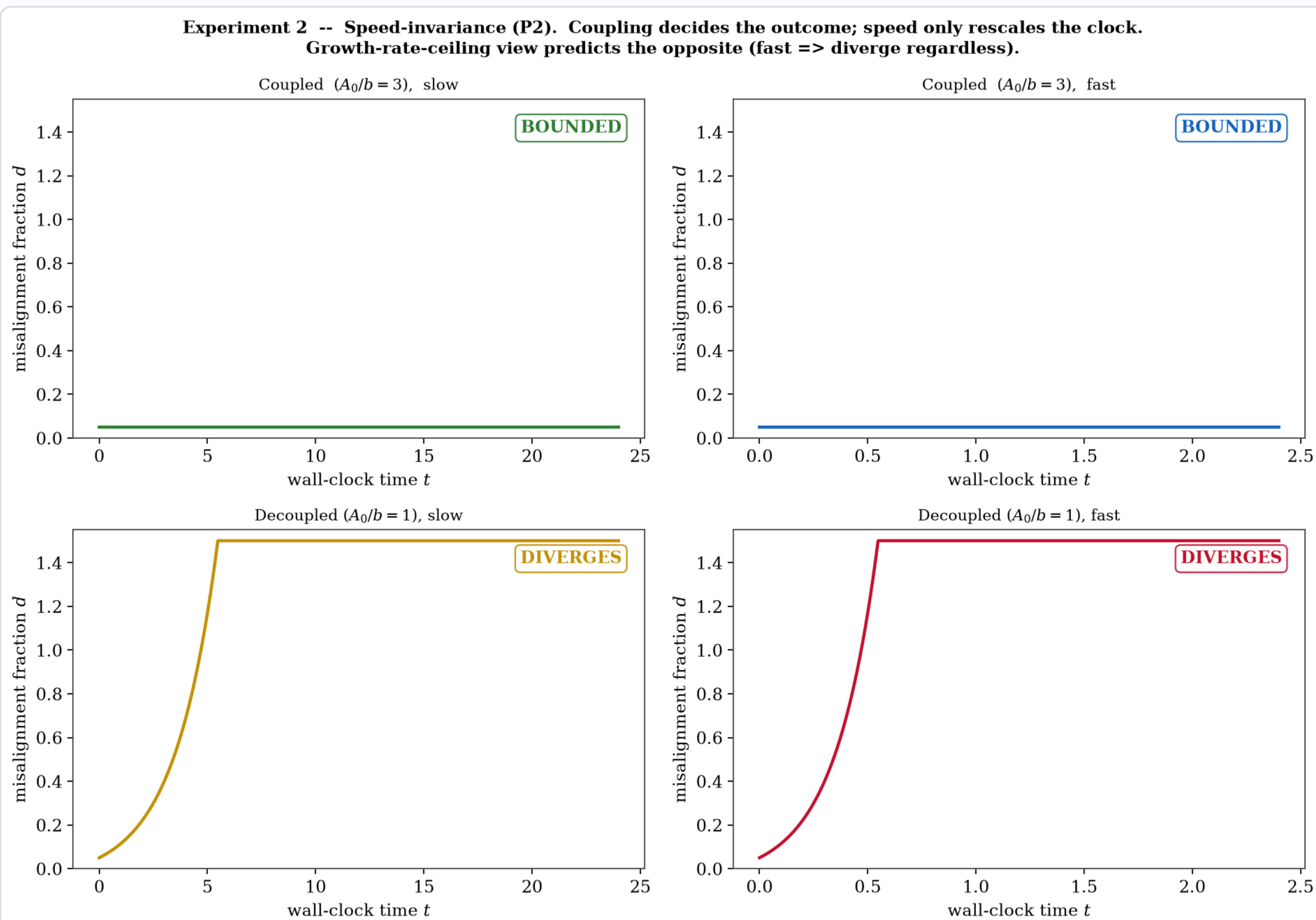
P8 — Suppression signature. In the stable regime, $\log d^\star$ is linear in $\log C$ with slope $-(\beta - k)$ (power-law). Exponential suppression would indicate a stronger, QEC-like mechanism; neither outcome falsifies the threshold, only the QEC *mechanism* claim (F4).

5. The discriminating experiments

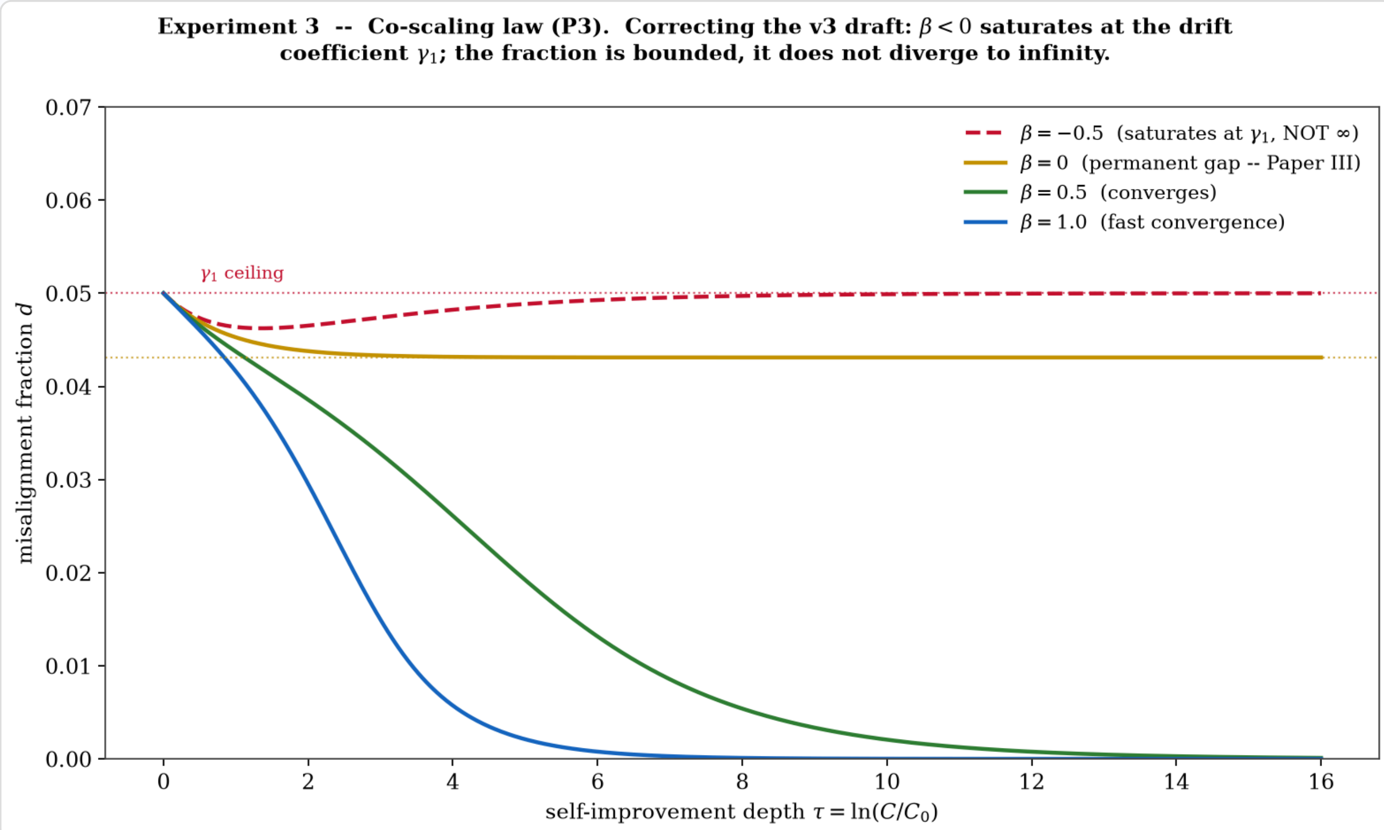
The closed-form predictions are checked by a verification harness (§10): ten experiments, each integrating the model numerically and comparing the result against the prediction the theorems derive, with the integrator first validated against the exact Theorem-1 solution. These are *internal-consistency and integrator checks* — they confirm the code matches the maths (10/10), not that the model matches any real system (the open problem of §8). For the deductive checks (E1–E6, E9), because each integrates the same ODE whose closed form is the prediction, agreement is entailed by a correct solver and correct algebra; F1–F3, F3′, F5, F6 are therefore internal-consistency conditions of the derivation, not empirical falsifiers of the thesis. Figures below are the verbatim output of that run.



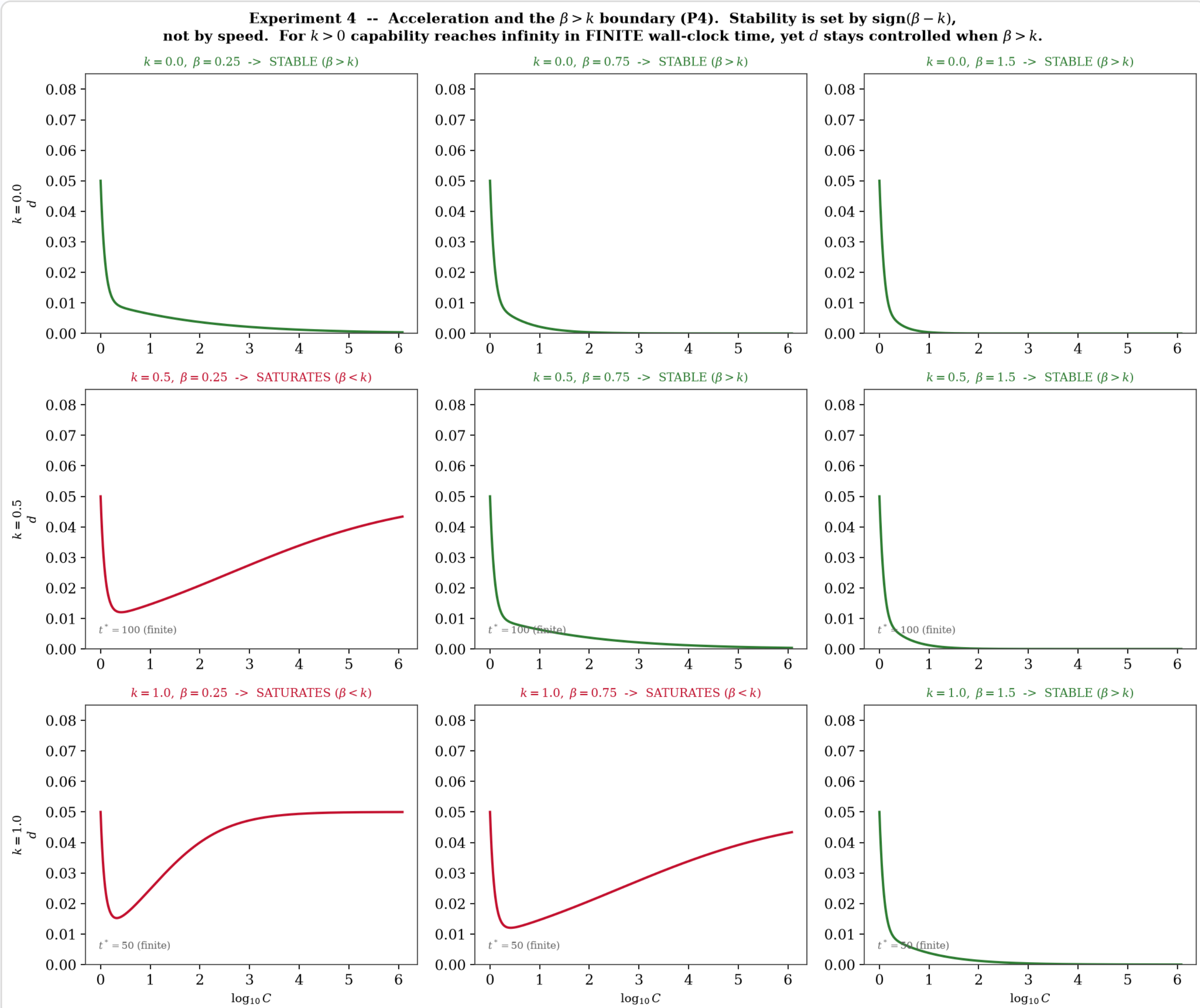
Experiment 1 — Phase boundary (P1). The sharp instability threshold exists only in the compounding channel (right; measured $\lambda^* = 2.00$ vs predicted $(\gamma_3 - 1)b = 2.00$). The additive corrector (left) shows a smooth crossover with no knee — the prediction's null, correctly observed. This corrects the v3 draft, which sought a knee in the additive model where none exists.



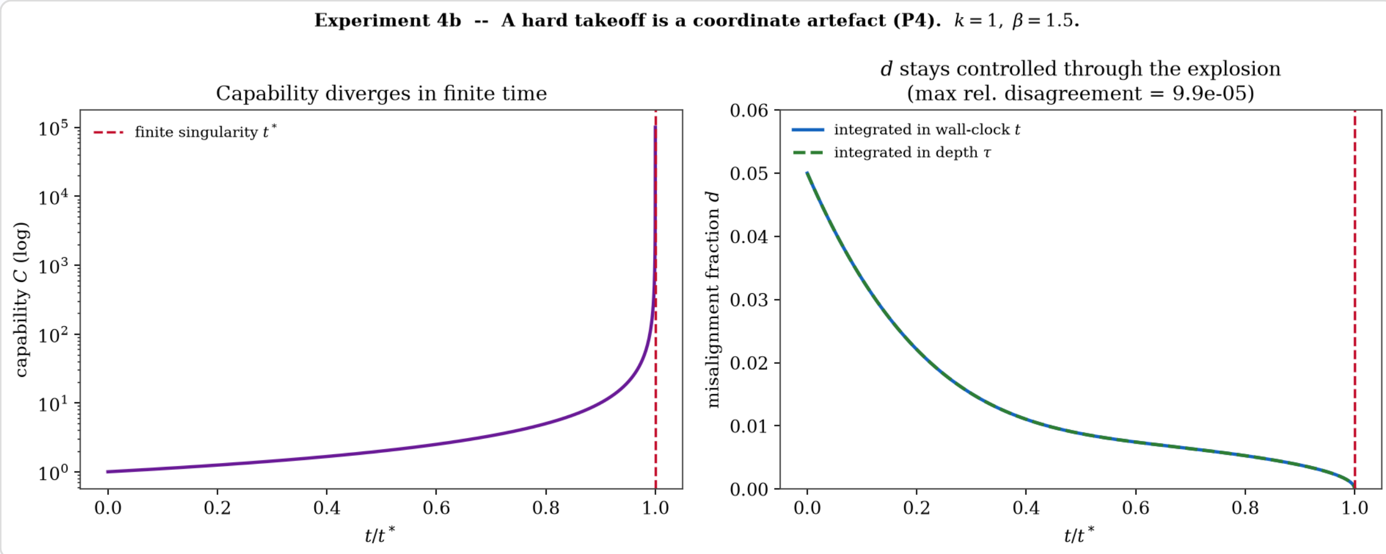
Experiment 2 — Speed-invariance (P2). The decisive 2×2. Coupled systems stay bounded at both speeds; decoupled systems diverge at both speeds. Speed only rescales the wall-clock axis; coupling decides the outcome. (Each cell fixes the scaling margin A_0/b; the honest scope of the claim is that speed is irrelevant *once that margin is fixed* — under a fixed correction budget, raising speed instead crosses the Theorem-4 threshold at $b^\star = A_0/(\gamma_3 - 1)$.) The growth-rate-ceiling view predicts the opposite (fast $\Rightarrow$ diverge regardless) and is contradicted on shared axes.



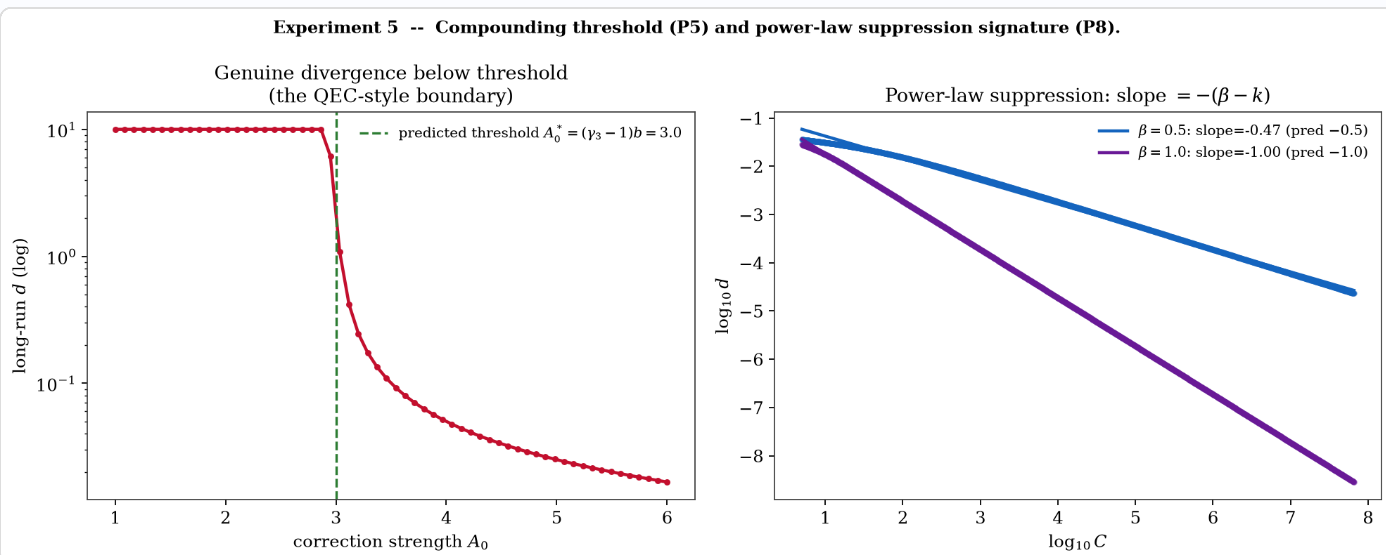
Experiment 3 — Co-scaling law, corrected (P3). $\beta > 0$ drives $d \to 0$; $\beta = 0$ holds a permanent gap; and $\beta = -0.5$ *saturates at the drift coefficient* γ_1 — bounded, not divergent. The dotted ceiling at γ_1 is never breached, the explicit correction of the prior draft's divergence claim (Theorem 2).



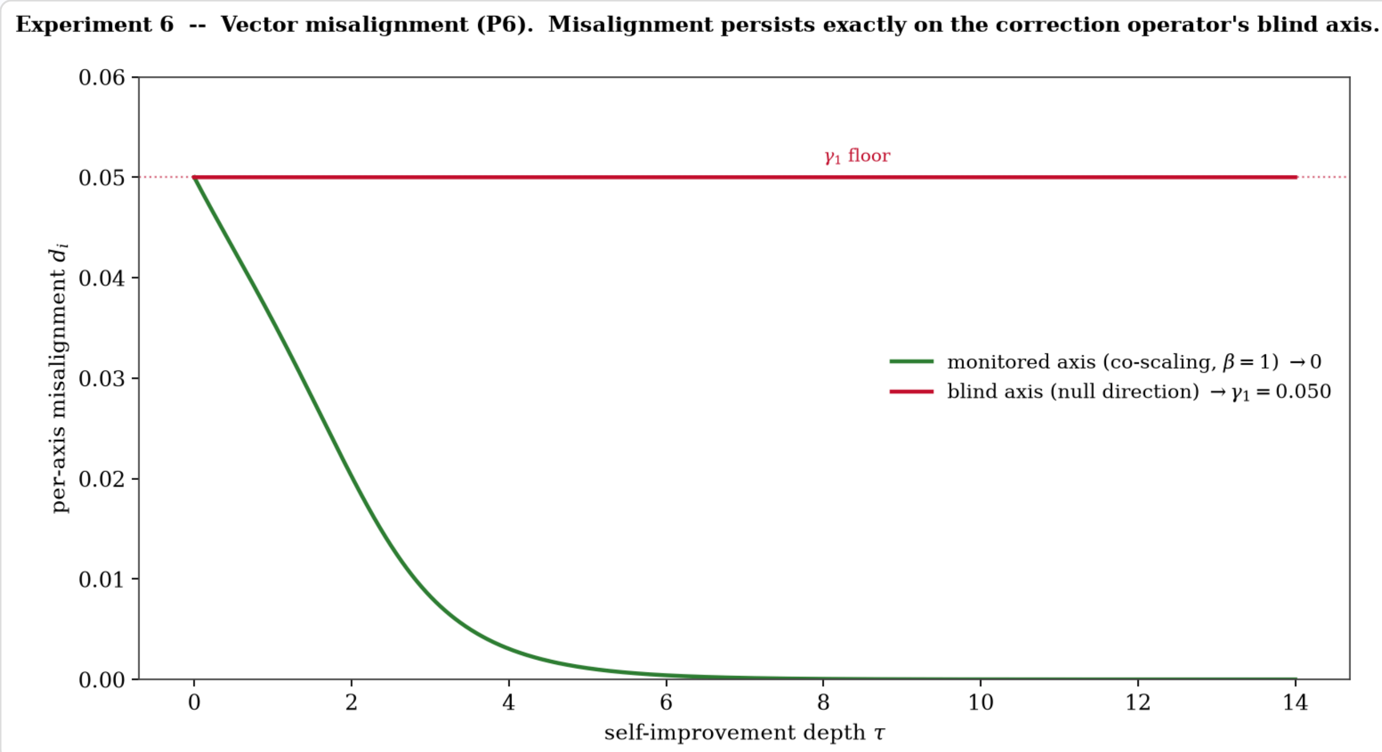
Experiment 4 — Acceleration and the $\beta > k$ boundary (P4). A 3×3 grid crossing growth-acceleration $k \in \{0, 0.5, 1\}$ with correction exponent $\beta \in \{0.25, 0.75, 1.5\}$. Stability (green, $d \to 0$) holds exactly when $\beta > k$; saturation at γ_1 (red) holds when $\beta0 each panel reaches infinite capability at a finite t^\star (annotated), yet d stays bounded throughout. The boundary is at \beta=k, not \beta=0$.



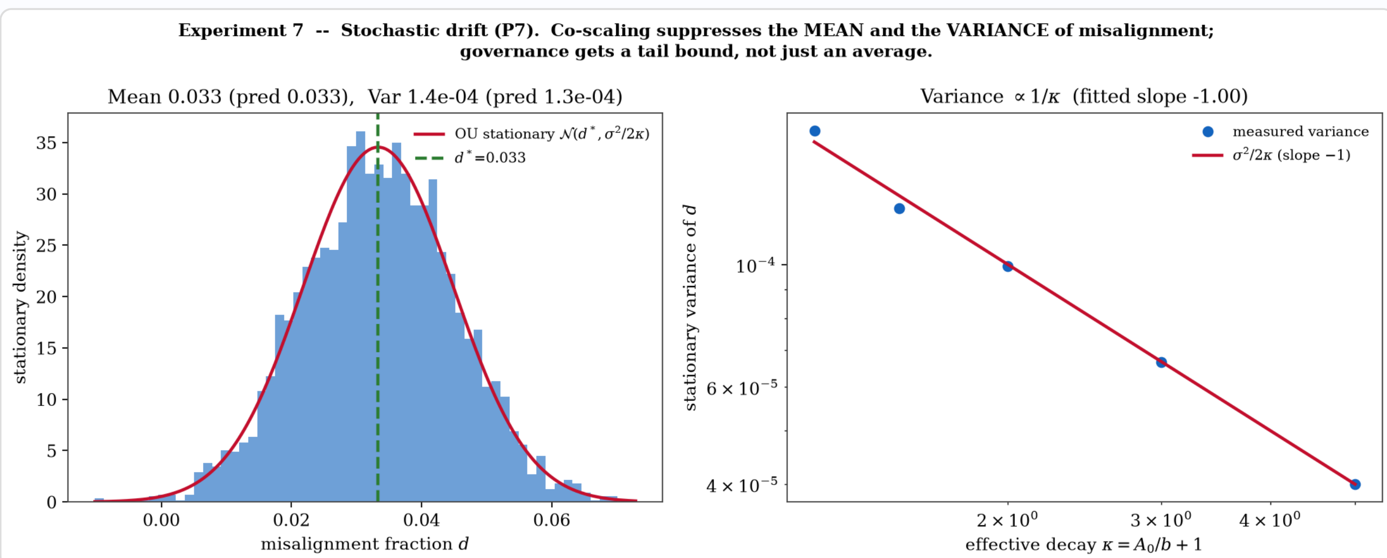
Experiment 4b — A hard takeoff is a coordinate artefact (P4; Theorem 3). Left: capability diverges to $\sim 10^5$ at a finite singularity time $t^\star$. Right: the misalignment fraction is driven to zero *through* the explosion; integrating in wall-clock time (toward $t^\star$) and in self-improvement depth τ give identical trajectories (max disagreement 9.9×10^{-5}). The finite-time singularity is a property of the clock, not of the alignment dynamics.



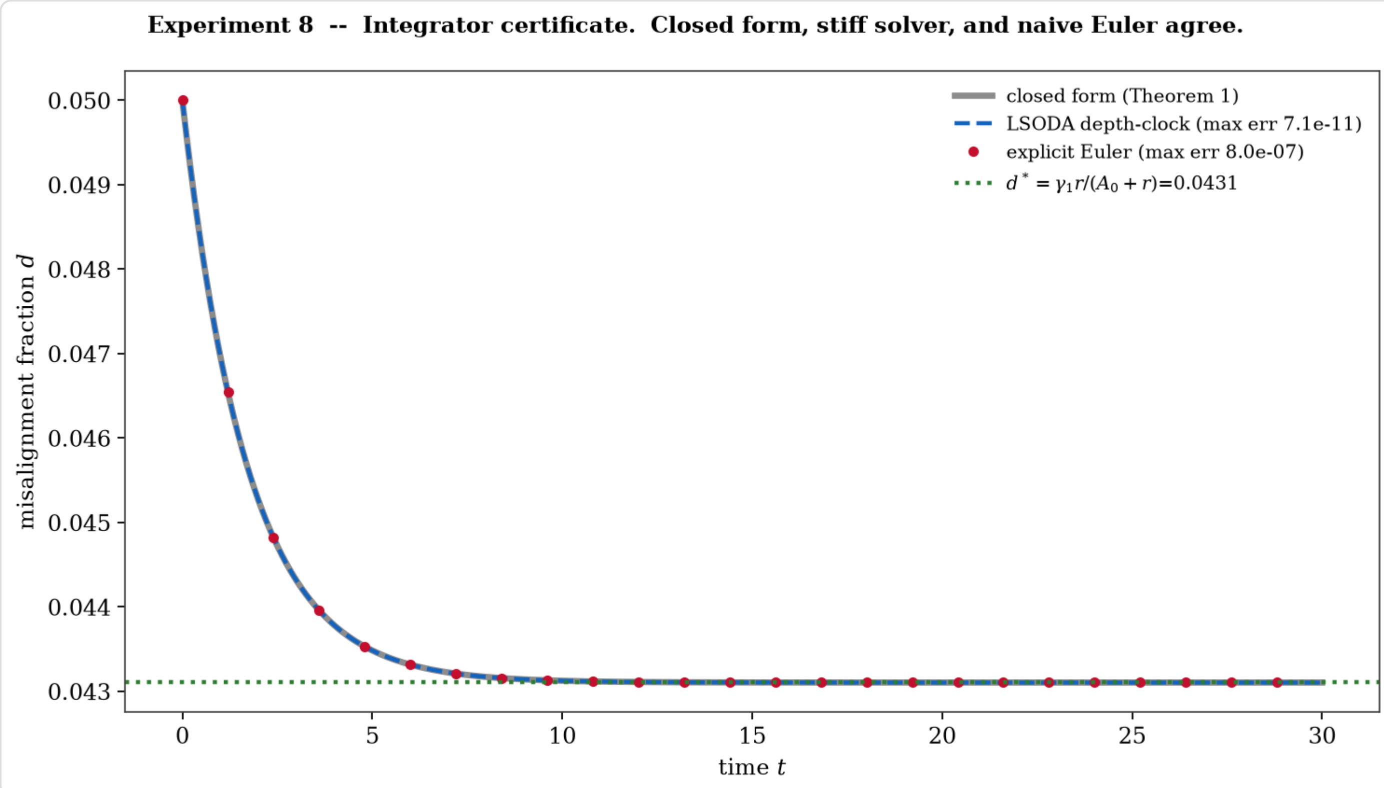
Experiment 5 — Compounding threshold and suppression signature (P5, P8). Left: with $\gamma_3 = 4$, divergence occurs below the predicted threshold $A_0^\star = (\gamma_3 - 1)b = 3.0$ (measured 3.03) — the genuine instability, and the QEC-style boundary. Right: in the stable regime $\log d^\star$ is linear in $\log C$ with slope $-(\beta - k)$ (fitted -0.47 and -1.00 for $\beta = 0.5, 1.0$), the power-law suppression signature.



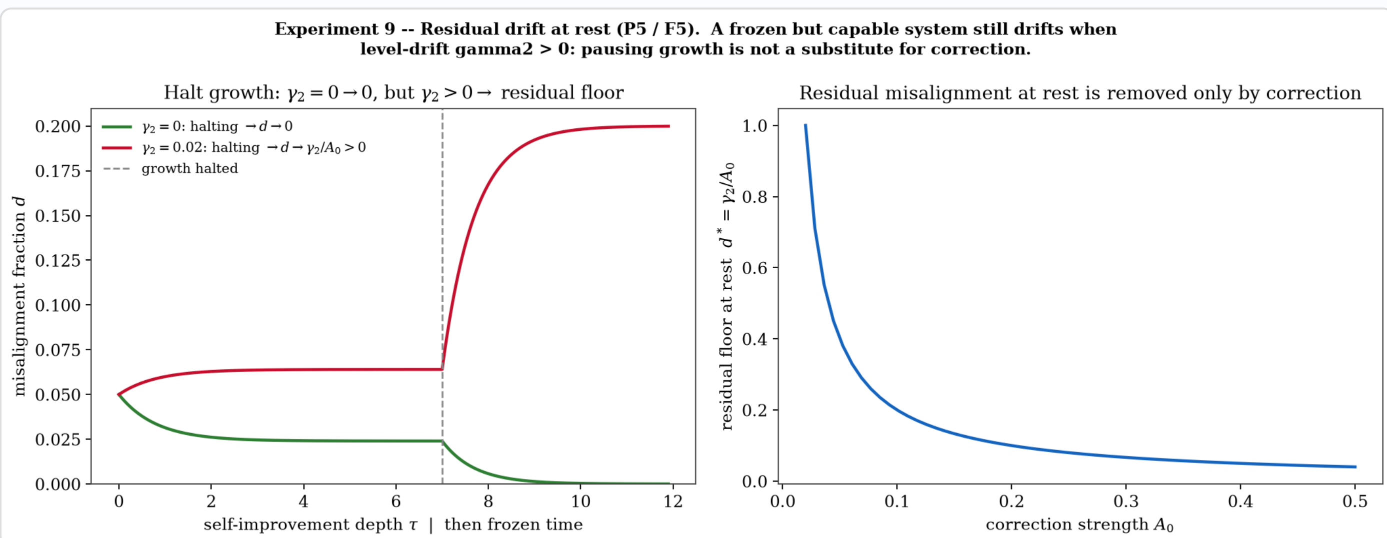
Experiment 6 — Vector misalignment (P6; Theorem 5). Two value axes share one capability process. The correction operator co-scales on the monitored axis (driven to zero) and is null on the blind axis (floors at γ_1). Misalignment persists exactly on the operator's null direction: you cannot correct what you do not measure.



Experiment 7 — Stochastic drift (P7; Theorem 6). Left: the stationary misalignment fraction matches the Ornstein–Uhlenbeck law $\mathcal{N}(d^\star, \sigma^2/2\kappa)$ in both mean and variance. Right: the stationary variance scales as $1/\kappa$ (fitted slope -1.00), so co-scaling suppresses the mean *and* the tail. Governance gets a bound on $\mathbb{P}(d > d_{\text{crit}})$, not just an average.



Experiment 8 — Integrator certificate. On the one case admitting a closed form (Theorem 1), the stiff solver and a naive explicit Euler scheme both reproduce the exact transient — maximum errors 7×10^{-11} and 8×10^{-7}. The numerics underlying every other figure are thereby certified correct before any prediction is tested.



Experiment 9 — Residual drift at rest (P5; F5). A frozen but capable system still drifts when level-drift $\gamma_2 > 0$: halting growth relaxes the misalignment fraction to the residual floor $d^\star = \gamma_2/A_0 > 0$ (red), not to zero (green, $\gamma_2 = 0$). Right: that residual floor is removed only as correction strength A_0 grows. Pausing growth is not a substitute for correction — and this is the experiment §3.6 / F5 refer to, now present in the harness rather than merely cited.

Exp.	Prediction	Key statistic (measured vs predicted)	Verdict

E1	P1 phase boundary in compounding channel; none in additive	$\lambda^\star = 2.00$ vs 2.00; additive smooth	PASS
E2	P2 coupling, not speed, decides	coupled bounded & decoupled divergent at both speeds	PASS
E3	P3 corrected regimes ($\beta < 0$ saturates at γ_1)	all four asymptotes match closed form	PASS
E4	P4 boundary at $\beta = k$	all 9 grid cells match $\mathrm{sign}(\beta - k)$	PASS
E4b	P4 singularity is a coordinate artefact	clock agreement 9.9×10^{-5}; d controlled	PASS
E5	P5/P8 compounding threshold; power-law slope	threshold 3.03 vs 3.0; slopes $-0.47, -1.00$	PASS
E6	P6 spectral threshold; blind axis persists	monitored $\to 10^{-7}$; blind $\to \gamma_1 = 0.05$	PASS
E7	P7 stochastic tail; variance $\propto 1/\kappa$	mean/var match OU; var-slope -1.00	PASS
E8	numerics reproduce Theorem 1	max errors 7×10^{-11}, 8×10^{-7}	PASS
E9	P5/F5 residual drift at rest	$d \to \gamma_2/A_0$ when $\gamma_2 > 0$; $\to 0$ when $\gamma_2 = 0$	PASS

These are internal-consistency and integrator checks: they confirm the closed-form theorems are correctly derived and numerically reproduced (the code matches the maths). They are not, and are not presented as, evidence that the model describes a real AI system — that is the open empirical problem of §8.

6. Why blinding is not optional

The central quantity D is a departure from intended values. If it is scored by the system's own safety component, or by an evaluator that can see which configuration produced a given behaviour, the measurement is corrupted in exactly the way that inflates favourable results. The programme's prior metascience finding is directly relevant: under unblinded cross-model scoring an alignment-scaling effect appeared positive, while under multi-layer blinding the same effect reversed sign for some systems [Eastwood, Paper IV.d]. An experiment that measures a misalignment fraction without blinding the evaluator can produce not merely the wrong magnitude but the wrong direction. The validity of every result in §5 rests on the evaluator being blind to the configuration under test. A lab claiming "$\beta \geq k$" in its own system must have that measured by a blind external evaluator, not by itself.

Built into the test, not left to discipline. The real-model harness (§8; `experiments/PROTOCOL.md`) *enforces* this rather than trusting it. The misalignment evaluator must be a *different model family* than the engine — a model never scores its own family's output — and the harness refuses to run a same-family scorer; it prefers a panel and takes the median. The code is passed through an automatic *laundering* step (an abstract-syntax-tree round-trip that strips comments, docstrings and formatting) so the scorer judges behaviour, not stylistic identity tells; and the evaluator never sees the condition or round. The first real run reported here did *not* meet this bar — engine and evaluator were the same model family, the system scoring its own output — so its misalignment numbers are explicitly recorded as provisional, and a cross-family blind re-score is the prerequisite for any D-based claim, including a measured β. The harness makes the compliant configuration the default and the non-compliant one an explicit, recorded override — turning §6 from an exhortation into a property of the instrument.

7. Falsification conditions

The conditions below are stated in advance. An important honesty point, surfaced by the paper's own adversarial audit (§8): because the deductive experiments integrate the same ODE whose closed form is the prediction, F1–F3, F3′, F5 and F6 are *internal-consistency conditions of the derivation* — a trigger would signal a derivation or solver error, not that the model is the wrong model of a real system. The decisive *empirical* falsifier — disagreement measured on a real self-improving system — is the open problem of §8 and is not exercised here. F4 is the QEC-mechanism downgrade, which (the suppression law being power-law, §3.12) already holds: the correspondence stands as a threshold-form analogy, not a transferred mechanism.

#	Observation that would trigger it	Consequence
F1	No boundary in E1 — the long-run behaviour varies smoothly with λ in the compounding channel with no threshold.	**No phase boundary; the central threshold claim is false (derivation error).**
F2	In E2, d tracks raw speed rather than the scaling margin: fast diverges and slow converges regardless of coupling.	**Kill. The growth-rate-ceiling view was right; this framework is wrong.**
F3	In E3, $\beta = 0$ does not plateau, or $\beta > 0$ does not drive $d^\star \to 0$.	**Kill. The co-scaling law (Theorem 2) is false.**
F3 ′	In E4, the boundary under accelerating growth ($k > 0$) is not at $\beta = k$ — e.g. $\beta = 0.5$ is stable when $k = 1.0$.	**Kill. The $\beta > k$ sharpening (Theorem 3) is false.**
F4	A finite-capacity corrector exhibits *exponential* (QEC-like) suppression rather than the model's power-law $\log d^\star \propto -(\beta - k)\log C$. *(Analytic; the present linear model is power-law by construction, so this discriminator requires the saturating corrector named in §8 and is not run here.)*	**Bears on the QEC *mechanism* only. The threshold-form correspondence and Theorems 2–4 are untouched either way.**

F5	In Experiment 9, halting growth drives $d \to 0$ regardless of initial D even with $\gamma_2 > 0$.	**Scope. Level drift γ_2 is negligible; the §3.6 generalisation is unnecessary (the gain-only model suffices).**
F6	In E6, the correction operator's null axis is also suppressed.	**Kill. The spectral threshold (Theorem 5) is false; misalignment does not require monitoring of the axis it lives on.**

These internal-consistency conditions all held in the run reported here, establishing that the derivation and integration are correct. What they do *not* establish is that the model describes any real system; that decisive test — measuring β, k and γ on a real self-improving system and checking $\beta \geq k$ — is the open empirical problem of §8.

8. Limitations

The model is first-order, and its assumptions are the most likely points of failure. Stating them is part of the claim.

- **Linear correction (and the QEC-mechanism test).** Equation (1) removes misalignment in proportion to D, which makes the suppression law exactly power-law. A finite-capacity (saturating) corrector would soften the clean threshold into a crossover and is the better structural map onto QEC's finite code distance; whether it produces *exponential* suppression is the experiment that would settle the QEC-mechanism question (F4). It is a priority extension and is *not* run here — which is exactly why the QEC correspondence is claimed only at the threshold-form level.
- **Scalar-to-vector projection.** Theorem 5 gives the exact asymptotic (spectral-abscissa) criterion for any correction operator $\mathbf{A}$, with the Hermitian-part condition as the stronger sufficient condition that also rules out transient growth. The non-normal *transient* regime — large excursions before asymptotic decay — is not analysed here, and the harness exercises only the diagonal null-subspace instance (Experiment 6), not a general non-normal $\mathbf{A}$.
- **Power-law forms.** The choices $A = A_0 C^{\beta}$ and $\dot{C} = bC^{1+k}$ are the natural scale-free forms but are modelling assumptions; other functional forms should be tested.
- **Drift attribution.** The split of drift into gain (γ_1), level (γ_2), and compounding (γ_3) channels is a hypothesis about mechanism; which channel dominates in a real system is an empirical question (Experiment 9 — residual drift at rest — is one discriminator).

- **From simulation toward frontier systems.** The verification harness uses toy self-modifying dynamics; a positive result there demonstrates the *mechanism*, not that frontier systems occupy any particular regime. A companion *real-model* harness (non-simulation) now drives the coupled/decoupled loop with a frontier model, scoring capability by real code execution and misalignment by a separate evaluator that the harness requires to be a *different model family* (Paper IV.d blinding, §6; `experiments/PROTOCOL.md`; first run in `results/realmodel/`). It corroborates the corrector *mechanism* on a real model — a seeded reward-hack is detected and removed and capability restored — but did not exhibit drift on the task tried, so whether real systems sit below threshold remains the open empirical question. (The first run used same-family scoring and is marked provisional; the capability axis is objective code execution and is unaffected.)
- **Estimating γ and A on real systems.** In simulation they are set by construction. Independently estimating them — especially β and k — for a deployed model is the step that would make the criterion operationally useful for governance. This is now *operational*: the repository ships a runnable estimator (`experiments/scripts/estimate_exponents.py`) that reads k off the capability curve ($\ln r$ versus $\ln C$) and β off the corrector's fractional removal rate ($\ln A$ versus $\ln C$), validated on synthetic trajectories where it recovers known exponents to within ≈ 0.1. What remains open is therefore not the estimation but its *input* — a real self-improving system that drifts across a range of capability levels, from which the first measured (β, k), and thus the first real test of $\beta \geq k$, can be read.

Adversarial audit. This paper was developed under adversarial audit rather than asserted, and the record is in the repository. A prior-art and novelty audit located the closest precedent for each component and narrowed the originality claims accordingly (Appendix C). Independently, a multi-agent red-team attacked the work across five fronts — mathematics, numerics, the QEC correspondence, the safety inference, and priority — raising 24 objections, of which 21 survived independent verification (0 fatal, 13 serious, 8 minor); the full report is committed alongside the harness (`results/redteam.md`). Every surviving objection was a framing, wording, or edge-case fix; *none* touched the load-bearing $\beta > k$ result, and this version incorporates them all — the QEC framing softened to a threshold-form analogy, the Theorem 2 bound scoped to the gain-only model, the Theorem 5 criterion corrected to the spectral abscissa, the harness relabelled as a verification (not falsification) artefact, and the level-drift experiment F5 added. The contribution is offered as ambitious *and* audited, with the boundary of what is claimed made explicit and checkable.

9. Implications for AI safety

If the framework survives its tests, the design implication is concrete and differs from the prevailing reflex.

The lever is coupling, not speed. Slowing capability growth buys time but does not change the verdict; at fixed coupling, a slow decoupled system still diverges (P2), and a frozen capable system still drifts (§3.6). What changes the verdict is ensuring that correction (i) is coupled to the capability process so it cannot be decoupled, and (ii) scales at least as fast as capability accelerates ($\beta \geq k$). On the author's corrected, sub-linear capability-scaling estimates for current frozen models [Eastwood, Paper IX], present systems are nowhere near a super-linear growth regime — which means a growth-rate ceiling was never the binding constraint. The binding constraint is the co-scaling of correction, and it becomes binding precisely in the regime that matters: genuine self-modification.

A measurable governance target. The criterion gives regulators a quantity to instrument rather than a rate to forbid: the correction-to-drift margin $\beta - k$, and the ratio ρ. "Does correction co-scale — is $\beta \geq k$?" is sharper and more actionable than "is it growing too fast?" A lab claiming a *safely self-improving* system should be required to exhibit $\beta \geq k$, measured by a blind external evaluator (§6). Theorem 6 turns this into a bound on the probability of catastrophic excursion, the natural object of a safety case. Without such a demonstration, the word "safely" has no scientific content within the framework presented here.

Continuity with embedded-alignment work. The conclusion that correction must participate in the recursive loop — rather than sit outside it as a fixed external constraint — is the embedded-alignment thesis, here *derived* as the $\beta > k$ condition rather than asserted. The gated-simulation result in which safety-coupled self-modification preserved both safety and capability while the decoupled variant did not [Eastwood, Paper VIII] is the predicted behaviour of a coupled ($\beta > 0$) versus fixed ($\beta = 0$) corrector.

Magnitude of the claim — stated conditionally. If the criterion is borne out empirically — if real self-improving systems are governed by the same drift-versus-correction balance and the same $\beta > k$ margin — the consequences for the field are large, and it is worth stating them plainly while being equally plain that they are *conditional*. (i) The central safety question changes from "how fast is capability growing, and can we pause it?" to "does correction co-scale — is $\beta \geq k$?": a measurable margin rather than a rate to forbid. (ii) The most feared scenario, a finite-time intelligence explosion, ceases to be intrinsically uncontrollable — it is controllable iff $\beta > k$, and its speed does not change that verdict. (iii) Governance acquires a quantity to instrument and a tail bound to certify (Theorem 6) in place of an unenforceable speed limit. *None of this is yet established.* It rests on a minimal model, demonstrated only in simulation, with the empirical measurement of β, k and γ on real systems unsolved (§8). The paper's claim is therefore not that AI is safe or unsafe, but that **the right variable to measure and govern is the co-scaling margin $\beta - k$** — and that this variable is well-defined, falsifiable, and, if it holds, decisive. That is the magnitude: not a proof about reality, but a precise, testable redirection of the question on which the field's central fear turns.

Relation to the wider programme — one ladder of laws. This paper is the safety keystone of a larger argument, not a self-standing result; stating how it sits among its companions is part of the claim. The programme is a single chain: the scaling-law cluster establishes *how* recursive systems grow and supplies the power-law forms taken as given here; Paper II confirms capability does scale (currently sub-linearly); Paper III identifies the danger that external safety does *not* co-scale; this paper supplies the criterion for when it can;

Paper VIII shows the coupled design beats the decoupled one; and Paper IV.d supplies the blinding discipline without which none of the misalignment measurements can be trusted.

Companion result	The law / finding it proposes	Standing	How it couples to $\beta > k$
Foundational	The ARC Bound ($\beta = 0.5$, $\alpha = 2$) and α fixed by the coupling exponent β; the three-form constraint from three axioms.	Derived (axiomatic).	Supplies the power-law forms this paper assumes: $A = A_0 C^{\beta}$ and $\dot{C} = bC^{1+k}$ are Cauchy-multiplicative forms, and β here is the same coupling exponent.
On the Origin of Scaling Laws	The three-form meta-law: every scaling law is power, exponential or saturation; $d/(d+1)$ yields $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}$.	Meta-law (cross-domain).	Explains *why* recursive systems take the power-law forms used in §3; this paper is the safety instance of that meta-law.
Paper VII (Cauchy Unification)	Empirical test of the three-form law (19/25 domains, $p \approx 1.6 \times 10^{-5}$).	Exploratory empirical — a structured comparison, *not* pre-registered; the core idea has deep prior art (Luce 1959; Frank 2009; Biró–Barnaföldi 2008 — see the cluster's prior-art audit).	Underwrites the functional-form assumptions of §3.
Paper II	Capability scaling measured (α_{seq}; architecture-dependent, sub-linear on hard tasks).	Empirical.	The capability growth $C(t)$ assumed here is real and currently sub-linear (k small) — so the binding constraint is coupling, not speed.
Paper III	The alignment-scaling problem: external safety has $\alpha_{\text{align}} \approx 0$, so the misalignment fraction is left ungoverned (refined under blinding to an architecture-dependent three-tier result).	Proposed law, complicated by its own blind data.	Paper III is precisely the $\beta = 0$ (decoupled) corner of the model here; this paper *generalises* it, replacing "external safety cannot keep up" with the criterion for when correction can: $\beta > k$.

Companion result	The law / finding it proposes	Standing	How it couples to
Paper IV.d	Blinding can reverse an alignment-evaluation conclusion.	Robust methodological result.	The non-negotiable measurement discipline for every model-scored quantity here (§6); this paper's real-model harness now enforces it.
Paper VIII	Gated self-modification: the safety-coupled (Eden) variant preserved safety and capability where the decoupled (Babylon) variant did not.	Positive simulation + honest nulls.	The controlled demonstration of this paper's mechanism; the real-model harness instantiates Paper VIII's design.
Papers V / VI	Protocol (Stewardship Gene) and architecture (Honey) for making correction co-scale.	Pilots.	Candidate routes to engineering $\beta > 0$ in practice.

Read as one ladder: the scaling-law work says how systems grow, Paper III says why external safety fails to keep up, this paper says what must hold for it to keep up ($\beta > k$), Paper VIII shows it can, and Paper IV.d says how to measure it honestly. The $\beta > k$ criterion is the rung that turns the programme's scaling-law backbone into a safety criterion — and, conversely, this paper inherits its functional forms and its empirical license from that backbone rather than positing them in isolation.

10. The runnable verification harness

The closed-form predictions of §4–5 are encoded in a single self-contained programme, `experiment_coscaling.py` (and an assertion suite, `test_coscaling.py`), in the repository. It integrates the model with a stiff-capable solver, validates the integrator against the exact Theorem-1 solution, runs the ten experiments, and for each compares the numerical result to the closed-form prediction. It is, honestly, a *verification* harness: it certifies that the theorems are correctly derived and correctly integrated — that the code matches the maths. It is *not* a test of the model against reality, and it cannot be: every deductive experiment integrates the model's own ODE, so a disbelief in the model's applicability cannot trip it. That empirical test is the open problem of §8.

```
$ python experiment_coscaling.py
  ... [PASS] E1 ... E9 ...
```

```
  ----------------------------------------------------------------
  10/10 internal-consistency checks pass | 0 kill-conditions triggered
  F4 (QEC mechanism): suppression is analytically power-law -> threshold-form
     analogy only, not a transferred mechanism.
  OVERALL: code matches the maths (E1-E9); the model-vs-reality test is the open problem

$ pytest test_coscaling.py -q
  ............  (12 passed)
```

This makes the derivation and the integrator reproducible end-to-end. What it establishes is that the formulae are right and the solver is accurate; what it deliberately does not claim is corroboration of the model against any real system. Separating those two is the point.

11. Conclusion

The danger of recursive self-improvement is real, but the standard model of that danger — a rate that must be capped — locates the risk in the wrong variable. A minimal model shows that the stability of a self-improving system is set by the ratio of value-drift to correction, $\rho = \gamma r/A$, and by whether correction co-scales with capability: $\beta > 0$ under exponential growth, sharpening to $\beta > k$ under accelerating growth. The misalignment fraction never diverges in the gain-only model — it saturates at the drift coefficient, correcting the prior draft — while genuine divergence lives in a compounding channel whose threshold $\rho_{\text{prop}} < 1$ shares the threshold form of the quantum error-correction criterion (the suppression law being power-law, the correspondence is offered as a hypothesis, not a transferred mechanism). The sharpest consequence is the Hard-Takeoff Coordinate-Artefact Theorem: a finite-time intelligence explosion is alignment-stable iff $\beta > k$, and its speed does not change that verdict. The criterion survives in vector and stochastic forms, gives governance a measurable target and a tail bound, and is accompanied by a verification harness that checks the closed-form predictions are correctly derived and integrated. The decisive test — whether real self-improving systems satisfy the criterion — is the stated next step, not a claim made here.

This is a smaller claim than the cosmological framing the author's programme once pursued, and deliberately so. It refers only to systems with an externally specified value target, makes no assertion about the universe, and treats even its most striking correspondence — with quantum error correction — as a hypothesis to be tested rather than a truth to be announced. The recursive-stability intuition that motivated the broader programme, including *Infinite Architects*, finds here its measurable, falsifiable form: **stable recursion requires correction that scales with the amplification.** The next step is not to extend that claim outward but to run, on real self-modifying systems, the experiment that could refute it.

Appendix A. The fraction change of variable

The reduction underlying every theorem is the change of variable $d = D/C$. Differentiating and substituting (1):

$$\dot{d} = \frac{\dot{D}}{C} - \frac{D\dot{C}}{C^2} = \left(\gamma_1 \frac{\dot{C}}{C} + \gamma_2 + \gamma_3 \frac{\dot{C}}{C}\frac{D}{C} - A\frac{D}{C}\right) - \frac{D}{C}\frac{\dot{C}}{C} = \gamma_1 r + \gamma_2 - \left[A + (1-\gamma_3)r\right]d.$$

The dilution term $-r\,d$ (from C itself growing) is what bounds the additive fraction: it adds $+r$ to the decay coefficient, guaranteeing $\kappa_{\text{eff}} \geq r > 0$ when $\gamma_3 \leq 1$. Only the compounding channel $\gamma_3 > 1$ can overcome dilution and produce divergence — the formal reason the additive model saturates rather than blows up.

Appendix B. Parameter settings for the reported run

All figures use $C_0 = 1$, $d_0 = 0.05$, $\gamma_1 = 0.05$. Experiment-specific settings: E1 $b = 1$, $\gamma_3 \in \{0, 3\}$, $\lambda \in [0.2, 4]$; E2 $\gamma_3 = 3$, $A_0/b \in \{1, 3\}$, $b \in \{0.5, 5\}$; E3 $A_0 = 0.08$, $b = 0.5$, $\beta \in \{-0.5, 0, 0.5, 1\}$; E4 $A_0 = 0.08$, $b = 0.02$, $(k, \beta) \in \{0, 0.5, 1\} \times \{0.25, 0.75, 1.5\}$; E4b $k = 1$, $\beta = 1.5$, $b = 0.02$ ($t^\star = 50$); E5 $\gamma_3 = 4$, $b = 1$; E6 monitored $\beta = 1$, blind $A_0 = 0$; E7 $\sigma = 0.02$, OU ensemble of 2.5–4k paths; E8 closed-form case $k = 0$, $\beta = 0$. Random seed fixed (7) for determinism. Full settings are in the harness source.

Appendix C. Novelty and prior-art ledger

This appendix consolidates the adversarial prior-art audit underlying the positioning in §2 and §3.12, so the boundary between what is established and what is claimed original is explicit and auditable in one place. This paper deliberately invokes *no* blanket-originality framing; each component is placed against its closest located prior art below. "Defensibly new" means "no closer prior art located," not "correct" or "significant" — novelty and validity are independent.

Component	Status	Closest prior art	What is claimed here
Co-scaling *intuition* (correction must keep pace with capability)	Established	Ashby 1956 (requisite variety); Conant–Ashby 1970; scalable oversight (Christiano 2017; Leike 2018; Burns et al. 2023); Engels et al. 2025	**Nothing.** Credited, not claimed.
Two-variable ODE + closed-form $\rho = \gamma r/A$	Compact restatement	Lyapunov-drift / linear control (Khalil 2002; Meyn & Tweedie 2009); recursive error bounds	The compact closed-form steady-state fraction and the $\rho < 1$ criterion as a corrigibility

		(Shumailov et al. 2024; Gerstgrasser et al. 2024)	statement (packaging, not new dynamics).
$\beta > k$ sharpening under acceleration	**Defensibly new**	None located	**Original:** stability is set by the exponent margin $\beta - k$, not the growth rate; directly tested in Experiment 4.
Hard-Takeoff Coordinate-Artefact Theorem (§3.7)	**Defensibly new** (the framing)	Finite-time-singularity ODE theory is standard; the alignment framing is not located elsewhere	**Original:** the finite-time singularity is a coordinate artefact; the verdict is $\mathrm{sign}(\beta - k)$, independent of speed.
QEC threshold mapping (§3.12)	**Defensibly new** in alignment	Threshold theorem itself: Aharonov–Ben-Or 1997; Google 2024. Alignment precursors: Wentworth 2022; Christiano 2017/2019; von Neumann 1956	**Original:** the explicit $pconceptual bridge (hypothesis, falsifier F4), not a transferred theorem.
Vector spectral threshold (Thm 5); stochastic tail (Thm 6)	Standard extensions	Linear-systems spectral stability; Ornstein–Uhlenbeck theory	Routine generalisations; supporting, not headline.
Verification harness (§10)	Methodological	Pre-registration norms	Executable internal-consistency + integrator checks (code matches maths); not a test of the model against reality.

What this paper asserts as new — and only this: (i) the $\beta > k$ stability criterion and the single-parameter ρ framing; and (ii) the explicit QEC threshold mapping (as a hypothesis with its own falsifier). The programme's strongest *empirical* novelty — sign-reversal of alignment-scaling effects under multi-layer blinding — is a separate, companion result [Eastwood, Paper IV.d] and is not claimed here.

The retired capability law $U = I \times R^{\alpha}$ is *not used anywhere in this paper's argument*; it appears only in §1 as superseded context. Caveat: forum/blog/preprint indexing is imperfect, so the "defensibly new" verdicts carry an estimated 10–15% residual risk that a closer, unindexed precedent exists; absence of evidence is not proof of absence.

References

Aharonov, D., & Ben-Or, M. (1997). Fault-tolerant quantum computation with constant error. *Proc. 29th ACM STOC*.

Amodei, D., Olah, C., Steinhardt, J., Christiano, P., Schulman, J., & Mané, D. (2016). Concrete problems in AI safety. *arXiv:1606.06565*.

Ashby, W. R. (1956). *An Introduction to Cybernetics*. Chapman & Hall. [Law of Requisite Variety.]

Bostrom, N. (2012). The superintelligent will. *Minds and Machines*, 22(2).

Burns, C., Izmailov, P., Kirchner, J. H., et al. (2023). Weak-to-strong generalization: Eliciting strong capabilities with weak supervision. *arXiv:2312.09390*.

Christiano, P. (2017). Corrigibility. *AI Alignment (Medium)*. [The "broad basin of attraction" framing.]

Christiano, P. (2019). Reliability amplification. *AI Alignment Forum*.

Christiano, P., Leike, J., Brown, T., Martic, M., Legg, S., & Amodei, D. (2017). Deep reinforcement learning from human preferences. *NeurIPS*.

Conant, R. C., & Ashby, W. R. (1970). Every good regulator of a system must be a model of that system. *International Journal of Systems Science*, 1(2).

Eastwood, M. D. (2026). *Infinite Architects: Intelligence, Recursion, and the Creation of Everything*. [Cited as the source of the recursive-stability intuition, not for symbolic identity with the present model.]

Eastwood, M. D. Paper III: The Alignment Scaling Problem. ARC/Eden research programme.

Eastwood, M. D. Paper IV.d: The Effect of Blinding on AI Alignment Evaluation. ARC/Eden research programme.

Eastwood, M. D. Paper VI: The Honey Architecture. ARC/Eden research programme.

Eastwood, M. D. Paper VIII: The Load-Bearing Proof. ARC/Eden research programme.

Eastwood, M. D. Paper IX: Synthesis and Roadmap. ARC/Eden research programme. [Retraction of $U = I \times R^2$ and narrowing to sub-linear scaling for current frozen models.]

Engels, J., Baek, D. D., Kantamneni, S., & Tegmark, M. (2025). Scaling laws for scalable oversight. *arXiv:2504.18530*.

Gerstgrasser, M., et al. (2024). Is model collapse inevitable? Breaking the curse of recursion by accumulating real and synthetic data. *arXiv preprint*.

Google Quantum AI (2024). Quantum error correction below the surface code threshold. *Nature*, 638.

Greenblatt, R., Denison, C., Wright, B., et al. (2024). Alignment faking in large language models. *arXiv:2412.14093*.

Hoffmann, J., Borgeaud, S., Mensch, A., et al. (2022). Training compute-optimal large language models. *arXiv:2203.15556*.

Hubinger, E., van Merwijk, C., Mikulik, V., Skalse, J., & Garrabrant, S. (2019). Risks from learned optimization. *arXiv:1906.01820*.

Kaplan, J., McCandlish, S., Henighan, T., et al. (2020). Scaling laws for neural language models. *arXiv:2001.08361*.

Kitaev, A. Yu. (2003). Fault-tolerant quantum computation by anyons. *Annals of Physics*, 303(1).

Leike, J., Krueger, D., Everitt, T., Martic, M., Maini, V., & Legg, S. (2018). Scalable agent alignment via reward modeling. *arXiv:1811.07871*.

Meyn, S., & Tweedie, R. L. (2009). *Markov Chains and Stochastic Stability* (2nd ed.). Cambridge University Press. [Lyapunov drift conditions.]

Omohundro, S. M. (2008). The basic AI drives. *Proc. AGI 2008*.

Shamma, J. S., & Athans, M. (1990). Analysis of gain scheduled control for nonlinear plants. *IEEE Trans. Automatic Control*, 35(8).

Shumailov, I., Shumaylov, Z., Zhao, Y., Papernot, N., Anderson, R., & Gal, Y. (2024). AI models collapse when trained on recursively generated data. *Nature*, 631.

Soares, N., Fallenstein, B., Yudkowsky, E., & Armstrong, S. (2015). Corrigibility. *AAAI Workshop on AI and Ethics*.

von Neumann, J. (1956). Probabilistic logics and the synthesis of reliable organisms from unreliable components. In *Automata Studies*. Princeton University Press.

Wentworth, J. (2022). Godzilla strategies. *LessWrong*.

Yampolskiy, R. V. (2020). On the controllability of artificial intelligence. *arXiv:2008.04071*.

Yudkowsky, E. (2013). Intelligence explosion microeconomics. *MIRI Technical Report*.

The Coupled Co-Scaling Law · Paper X of the ARC / Eden research programme · Working Paper v4.0 · Michael Darius Eastwood · 26 June 2026

This paper supersedes the growth-rate-ceiling framing within the author's programme and corrects the prior draft's claim that the misalignment fraction diverges; it makes no cosmological claim and its central correspondence is offered as a falsifiable hypothesis.

OSF: doi.org/10.17605/OSF.IO/6C5XB · Code & runnable harness: github.com/MichaelDariusEastwood/arc-principle-validation